

Solve problems involving the volume of right prisms including rectangular and triangular prisms, using established formulas and appropriate units

Learning intention

- solve problems involving the volume of right prisms including rectangular and triangular prisms, using established formulas and appropriate units.

Teach and model

- using dynamic geometry software, spatial reasoning and prediction to derive the formula for the volume of prisms.
- connecting the area of the floor space and the number of floors of a high-rise building to calculate the volume of a building.
- developing the connection between the area of the parallel cross-section (base), the height and volume of a rectangular or triangular prism to other prisms.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.